9 **(a)** **(i)** $E = hf = hc / \lambda$, so $E\lambda = hc$ with correct substitution of h and c M1

show conversion of E to eV and λ to nm correctly C1

$E\lambda = (6.63 \times 10^{-34})\ (3.00 \times 10^{8})\ (10^{9}) / 1.60 \times 10^{-19}$

$E\lambda = 1243$ eV nm A0

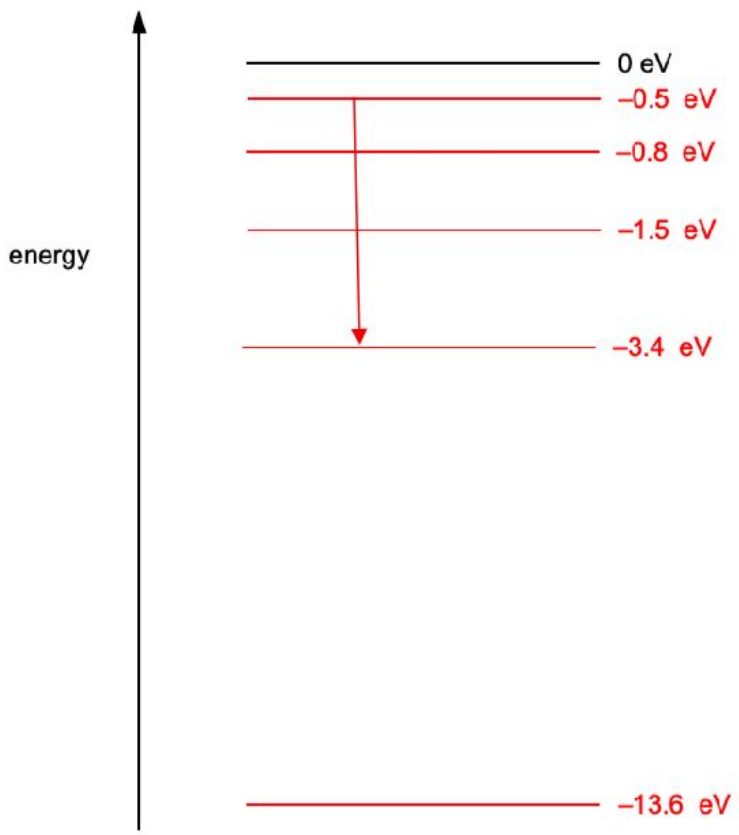
(ii) photon energy of 95.0 nm = 1243 / 95.0 = 13.1 eV A1

photon energy of 97.3 nm = 1243 / 97.3 = 12.8 eV A1

(iii) lowest energy at –13.6 eV B1

four energy levels (–0.5 eV, –0.8 eV, –1.5 eV and –3.4 eV) between 0 and –13.6 eV, deduct 1 mark for each missing level B3

increasing spacing as energy levels becomes lower B1



(iv) E = 1243 / 430 = 2.89 eV
arrow from –0.5 eV to –3.4 eV B1

(b) **(i)** wavelength of $\psi_B < \psi_A$ B1

since energy proportional to (momentum)^2 and momentum inversely proportional to wavelength (of electron) B1

so ψ_B has higher energy B1

(ii) electron transit from $n = 3$ (ψ_B) to $n = 1$ (ψ_A) C1

photon energy $= E_3 - E_1 = \frac{h^2}{8mL^2}(n_3^2 - n_1^2) = \frac{(6.63\times10^{-34})^2}{8(9.11\times10^{-31})(1.0\times10^{-10})^2}(3^2 - 1^2)$ C1

photon energy = 4.8 (or 4.82, 4.83) × 10^{-17} J A1

(c) **(i)** $\Delta p_y = h / \Delta y = 6.63 \times 10^{-34} / 1.5 \times 10^{-10}$ M1

$\Delta p_y = 4.4 \times 10^{-24}$ N s or 4.42×10^{-24} Ns A1

(ii) *y*-momentum uncertainty gives electrons a momentum (or velocity) perpendicular to the original direction (or stte) M1

(random process, so beam spreads out) with some electrons going to +*y* and some to –*y* (or stte) A1